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Paleomagnetism of Late Jurassic to Early Cretaceous red beds from the Cardamom Mountains, southwestern Cambodia: Tectonic deformation of the Indochina Peninsula

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ABSTRACT

Late Jurassic to Early Cretaceous red beds of the Phuquoc Formation were sampled at 33 sites from the Sihanoukville and Koah Kong areas of the Phuquoc–Kampot Som Basin, southwestern Cambodia. Two high-temperature remanent components with unblocking temperature ranging 650°–670 °C and 670–690 °C were identified. The magnetization direction for the former component ($D = 5.2°$, $I = 18.5°$ with $\alpha_{95} = 3.1°$ in situ) reveals a negative fold test that indicates a post-folding secondary nature. However, the latter component, carried by specular hematite, is recognized as a primary remanent magnetization. A tilt-corrected mean direction of $D = 43.4°$, $I = 31.9°$ ($\alpha_{95} = 3.6°$) was calculated for the primary component at 11 sites, corresponding to a paleopole of 47.7°N, 178.9°E ($A_{95} = 3.6°$). When compared with the 130 Ma East Asian pole, a southward displacement of $6.0° \pm 3.5°$ and a clockwise rotation of $33.1° \pm 4.0°$ of the Phuquoc–Kampot Som Basin (as a part of the Indochina Block) with respect to East Asia were estimated. This estimate of the clockwise rotation is ~15° larger than that of the Khorat Basin, which we attribute to dextral motion along the Wang Chao Fault since the mid-Oligocene. The comparison of the herein estimated clockwise rotation with the counter-clockwise rotation reported from the Da Lat area in Vietnam suggests the occurrence of a differential tectonic rotation in the southern tip of the Indochina Block. During the southward displacement of the Indochina Block, the non-rigid lithosphere under its southern tip moved heterogeneously, while the rigid lithosphere under the Khorat Basin moved homogeneously.

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1. Introduction

Following an initial model of extrusion tectonics in the Asian Continent due to the Indian Continent indentation discussed by Tapponnier et al. (1982), several researchers have reported a clockwise (CW) rotation accompanied by a southward displacement of the tectonic blocks located to the southeast of the Eastern Himalayan syntaxis (Peltzer and Tapponnier, 1988; Houseman et al., 1993; Royden et al., 1997). The present day occurrence of a CW tectonic rotation around the eastern Himalayan syntaxis has also been observed using GPS measurements (Gan et al., 2007; Allmendinger et al., 2007). A historical record of the CW rotation and southward displacement of the Indochina Peninsula has been documented using paleomagnetic and geological observations (Fig. 1). A thick succession of Jurassic to early Cenozoic continental red beds covers a wide area in the Indochina Peninsula (e.g., Fyhn et al., 2010; Morley, 2012). These red beds are distributed in a zone from the Lanping-Simao Basin in western Yunnan, through the Sayabouri-Nakhon Thai Basin in northern Thailand to the Khorat Basin in northeast Thailand. The reported paleomagnetic results from these rocks reveal the occurrence of a CW rotation together with a southward displacement of the Indochina Peninsula since the Cretaceous (Funahara et al., 1993; Yang and Besse, 1993; Sato et al., 1999, 2007; Charusiri et al., 2006; Tanaka et al., 2008; Takemoto et al., 2009; Otofuji et al., 2010, 2012; Tong et al., 2013; Singsoupho et al., 2014; Kornfeld et al., 2014a, 2014b). Geologi-

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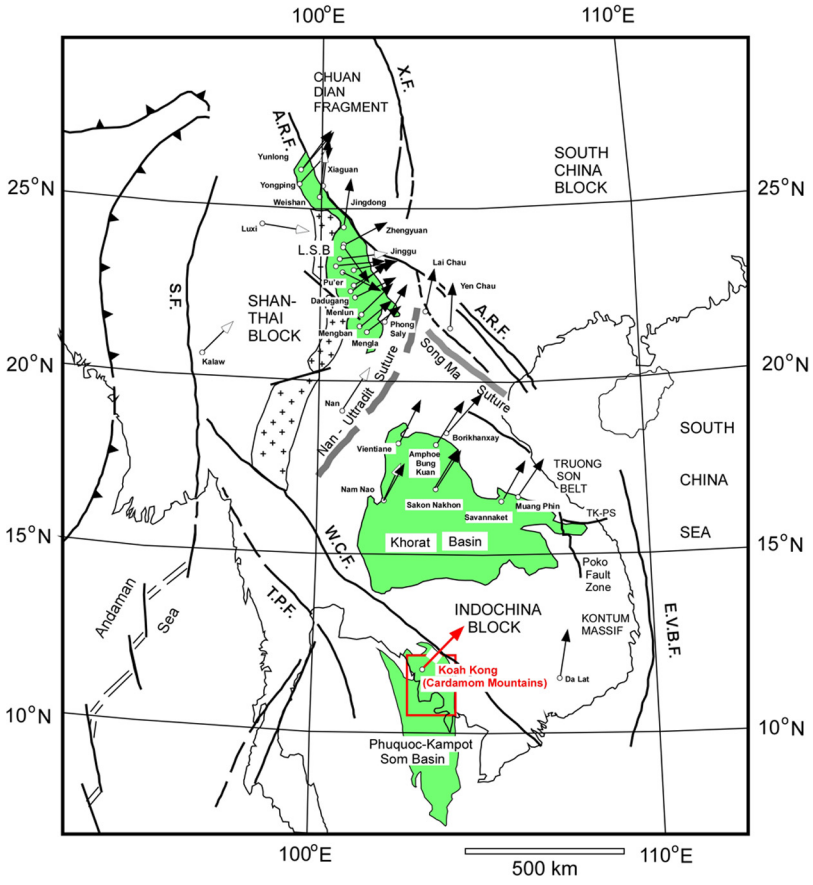


Fig. 1. Sketch map of the structure of the Indochina Peninsula and neighboring areas (modified from Otofuji et al., 2012). The green colored zones on the map indicate the distribution of the Jurassic and Cretaceous red beds. The arrows indicate the observed declinations of the Late Jurassic to Cretaceous rocks of the Indochina Peninsula, including areas from the Shan-Thai and Indochina blocks (Otofuji et al., 2012). The closed (open) arrows indicate data from the Late Cretaceous. A.R.F., Ailao Shan–Red River Fault; E.V.B.F., East Vietnam Boundary Fault; L.S.B, Lanping-Simao Basin; S.F., Sagaing Fault; T.P.F., Three Pagodas Fault; TK-PS, Tam Ky-Phuoc Son Shear Zone; W.C.F., Wang Chao Fault; X.F., Xianshuihe Fault. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

cal observations from the same area provide information about a lateral movement through the Red River Fault Zone, along which the Indochina Peninsula extruded up to 700 km (Leloup et al., 1995). On the basis of geo-chronological studies some researchers (Lacassin et al., 1997; Gilley et al., 2003) have assigned an age of 17–32 Ma to the sinistral slip movement along the Red River Fault.

Recent paleomagnetic studies (Kondo et al., 2012; Otofuji et al., 2012) of the Jurassic to early Cenozoic continental red beds not only predict a CW rotation of the Indochina Peninsula as a whole but its internal deformation as well. Although oroclinal bending was discovered in the Simao basin (Kondo et al., 2012), a counter-clockwise rotation (CCW) with respect to other basins in the Indochina Block was detected in the Da Lat basin of southern Vietnam (southern tip of Indochina) (Otofuji et al., 2012). A differential tectonic rotation within the Indochina Peninsula has also been claimed based on recent GPS measurements (Iwakuni et al., 2004; Shen et al., 2005; Calais et al., 2006; Gan et al., 2007; Simon et al., 2007). Further evidence regarding the internal deformation of the Indochina Peninsula is thus required to shed light on the strength of the East Asian lithosphere.

Bearing in mind the importance of further investigation, the red beds of the Phuquoc–Kampot Som Basin were chosen for a new paleomagnetic study. The basin is located to the south of the Khorat Basin and occupies a central part of the Indochina Block. According to some researchers (Lacassin et al., 1997; Fyhn et al., 2010; Morley, 2012), these two basins remained a part of a continuous large basin prior to its splitting by left-lateral tectonic movements along the Wang Chao Fault (Mae-Ping shear zone). To document the tectonic movement in the southernmost region of the Indochina Peninsula with respect to its main northern part, we sampled late Jurassic to early Cretaceous red beds from the Sihanoukville (11.5°N, 103.7°E) and Koah Kong (11.6°N, 103.5°E) areas in southwestern Cambodia, southwest of the Cardamom Mountains.

2. Geological setting and sampling

The Indochina Peninsula comprises a collage of continental blocks that include the Lanping-Simao Terrane and Indochina Block (Ferrari et al., 2008; Metcalfe, 2011). A widespread presence of Late Jurassic to Cretaceous sedimentary rocks characterizes the Mesozoic strata in the blocks. These non-marine layered rocks are mostly distributed in the Lanping-Simao Basin of southern Yunnan province and in the Khorat Basin of northeast Thailand.

A thick succession of sandstones and conglomerates of the Mesozoic Era is also distributed in the southwestern part of Cambodia. Their deposition occurred in a continental environment (Vysotsky et al., 1994; Ridd and Morley, 2011). The present day Cardamom Mountains in the Phuquoc–Kampot Som Basin (also called the Chuor Phnum Kravanh) are the main area of the deposition, where a succession on the order of 1000–2000 m has been recorded and named the Phuquoc Formation. In some reports, this succession has been named the Upper Sandstones or Grés Supérieurs (Bureau de Recherches Géologiques et Minières, 1971a, 1971b; Tien et al., 1991; Vysotsky et al., 1994).

While some geologists have assigned an age of Middle Jurassic-Early Cretaceous to the Phuquoc Formation (Bureau de Recherches

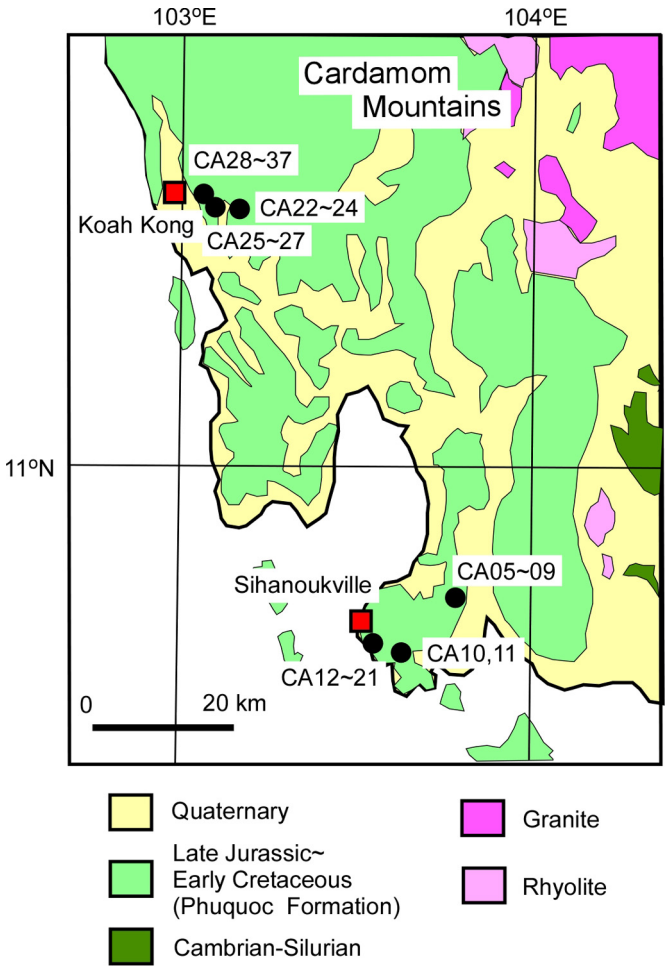


Fig. 2. Simplified geological map of the study area (after Vysotsky et al., 1994). The locations of the sampling sites are marked by closed circles.

Géologiques et Minières, 1971a, 1971b; Vysotsky et al., 1994), others have assigned it a Late Jurassic–Cretaceous age. On the basis of pollen species in the top layer of the succession, a Lower Cretaceous age was assigned to this formation by Workman (1977). Tansuwan (1998) has studied this formation in southeastern Thailand (western edge of the Phuquoc–Kampot Som Basin) and linked it to the Phu Kradung and Phra Wihan formations of the Khorat Plateau. A photogeological interpretation of a Google Earth terrain map (Ridd and Morley, 2011) found close similarities in the outlines of the Grés Supérieurs and Khorat Group. Folding with shallow dips is widespread in the area, and over a large area the bedding attitude is apparently subhorizontal.

As shown in Fig. 2, red sandstones of Late Jurassic to Early Cretaceous age in the Phuquoc Formation were sampled at 17 sites (CA05–CA21) from the Sihanoukville area (10.5°N, 103.7°E) and at 16 sites (CA22–CA37) from the Koah Kong area (11.6°N, 103.1°E). The outcrops at the sampling sites show a variation in dip from 1° to 29°. In general, 8 block samples were collected from 3 to 14 m covered area at each site. The orientations of the collected samples were determined using a magnetic compass. The present day geomagnetic field declinations at the sampling sites were obtained from the International Geomagnetic Reference Field "IGRF" (International Association of Geomagnetism and Aeronomy Working Group V-MOD, 2010).

3. Paleomagnetism

3.1. Laboratory procedures

Cylindrical samples of 2.5-cm diameter were drilled from every hand sample and then cut into 2.2 cm long specimens in the laboratory. Because of their fragile nature cylindrical specimens could not be cored from the samples taken at four of the sites (CA07, CA18, CA35 and CA36). The natural remanent magnetizations (NRMs) of the selected specimens were measured in a magnetically shielded laboratory at Kyoto University using a 2-G Enterprises three-axis cryogenic magnetometer (Pacific Grove, CA, USA). To monitor any possible chemical changes in the magnetic mineralogy during the thermal demagnetization (ThD), the magnetic susceptibility was measured after each heating step using a Bartington MS2 susceptibility meter. Thermal demagnetization (ThD) was performed using two thermal demagnetizers composed of non-inductively wound electric furnaces in three- or five-layer μ-metal magnetic shields. The internal stray field was less than 8 nT at the sample-heated position in each demagnetizer. All of the specimens were subjected to progressive ThD in 16–17 heating steps up to a maximum temperature of 690 °C. The temperature interval of the ThD was 50 °C from 100 to 500 °C (9 steps), 30 °C from 500 to 650 °C (5 steps) and 10–20 °C from 650 to 690 °C (2–3 steps). The demagnetization results were plotted on orthogonal vector diagrams (Zijderveld, 1967) and on equal-area projections. The paleomagnetic directions were determined by principal component analysis (Kirschvink, 1980). The Site-mean and formation-mean directions were calculated using Fisherian statistics (Fisher, 1953).

3.2. Thermal demagnetization

The NRMs of 128 samples from the Sihanoukville area and 109 samples from Koah Kong area were measured, and they revealed a wide range of intensities between 2.2×10^{-4} and 4.3×10^{-1} A/m. An abrupt decrease in the NRM intensities was observed at 100 °C from the ThD analyses for 83 samples (13 sites) taken from the Sihanoukville area and 8 samples (1 site) from the Koah Kong area, which indicates the presence of goethite (Fig. 3d, e, g). After the 100 °C demagnetization, the minimum and maximum NRM intensities were reduced to 1.8×10^{-4} and 4.8×10^{-2} A/m, respectively. The NRM intensities measured at 100 °C generally were greater for the Sihanoukville area samples than for those from the Koah Kong area; the mean intensities were $7.3 \pm 5.2 \times 10^{-3}$ A/m and $5.9 \pm 2.9 \times 10^{-3}$ A/m, respectively.

After the removal of the goethite carried component using the ThD procedure, single, two and three-component behaviors were observed in 56, 53 and 5 samples from the Sihanoukville area, respectively (Fig. 3). As shown in Figs. 3c and 3e, the stable single component magnetization was unblocked at 650 °C. In the case of samples with two-component behavior (Figs. 3b, d), the low-temperature component was unblocked at 300–350 °C or 560–590 °C, and the high-temperature component was unblocked by 650 °C (Fig. 3b, c). In the case of samples with three-component magnetization, the low, middle and high-temperature components were unblocked at 300–350 °C, 590 °C and 650 °C, respectively (Fig. 3a). The remaining 14 samples from the Sihanoukville area (sites CA05, CA06, CA10, CA16, CA20 and CA21) showed very unstable demagnetization behaviors.

The NRMs of the 109 samples from the Koah Kong area also showed three types of behavior after the 100 °C thermal demagnetization. Among the studied samples, single, two and three-component behaviors were observed in 52, 48 and 3 of the samples, respectively. An unblocking temperature of 670–690 °C, which

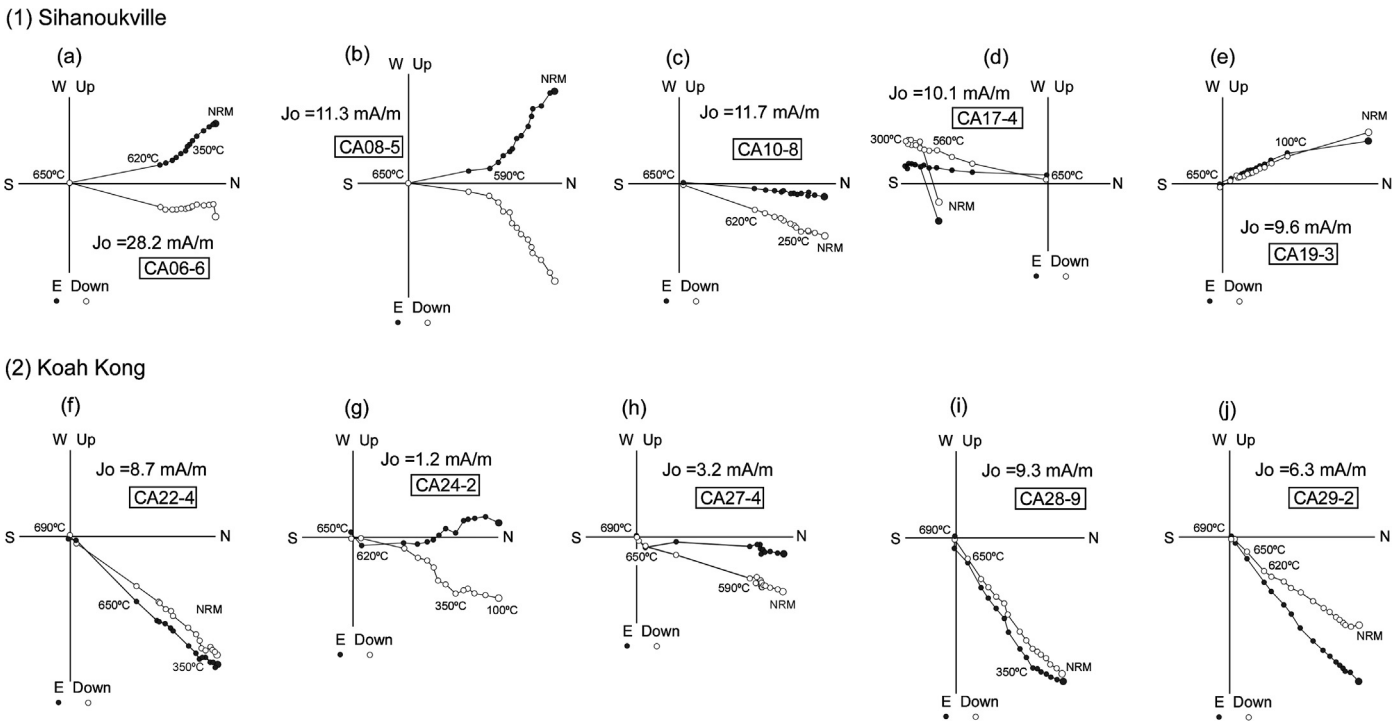


Fig. 3. Thermal demagnetization results on Zijderveld plots for the samples from the (1) Sihanoukville and (2) Koah Kong areas. The open (solid) symbols are the projections onto the vertical (horizontal) plane. All of the directions are plotted in geographic coordinates.

represents the high-temperature component, was exclusively detected in the samples from 10 sites (Fig. 3f, h, i, j). However, an unblocking temperature of 650 °C was only observed in the samples from sites CA24, CA25 and CA26 (Fig. 3g). Samples from site CA27 revealed two types of behavior (Fig. 3h); a single magnetization component with an unblocking temperature of 650 °C was observed in 2 samples, and two magnetization components with unblocking temperatures of 620 °C and 690 °C were observed in the remaining 8 samples. Because of their very unstable behavior, 4 samples from sites CA23, CA24 and CA26 were discarded.

Overall, two high-temperature unblocking ranges of 670–690 °C and 650–670 °C were identified in the analyzed samples, which we categorized as Type-A (Fig. 4) and Type-B (Fig. 5), respectively. Samples in the Type-A category exclusively belonged to 10 sites from the Koah Kong area, whereas those in Type-B belonged to 15 sites from the Sihanoukville area and 3 sites from the Koah Kong area. Site CA27 in the Koah Kong area produced two groups of samples, one with two unblocking temperatures (620 °C and 690 °C) and another with a single unblocking temperature (650 °C). As shown in Fig. 6, samples with unblocking temperatures of 620–650 °C and 690 °C were categorized as Type-A+B.

3.3. The NRM directions

The NRM directions of the studied samples depended on their unblocking temperatures (Fig. 7). The samples in the Type-A category were characterized by easterly deflected declinations and those in the Type-B by northerly declinations. The site mean directions for both of the categories were clearly divided into two groups (Table 1). In the case of site CA27, which yielded Type-A+B and Type-B samples, two site mean directions were calculated. One of them (Type A) was calculated from the remanent directions that had an unblocking temperature of 690 °C (Type-A in Type-A+B), whereas the second one (Type B) was obtained from the remanent directions that had an unblocking temperature of 620–650 °C (Type B and Type B in Type-A+B).

3.3.1. The NRM directions of the Type-A samples

Type-A samples with normal polarity inclinations and easterly deflected declinations were exclusively observed in 11 sites from the Koah Kong area. The site mean directions for those sites were tightly clustered, as evidenced from the precision parameters (larger than 13.6). The formation mean directions calculated on the basis of the 11 sites were $D_g = 42.6°$, $I_g = 30.8°$, $k_g = 162.0$ and $\alpha_{95} = 3.6°$ in geographic coordinates and $D_s = 43.4°$, $I_s = 31.9°$, $k_s = 158.7$ and $\alpha_{95} = 3.6°$ in stratigraphic coordinates (Fig. 7); they deviate significantly from the present day IGRF direction ($D = 359.5°$, $I = 9.5°$) and the axial geocentric dipole direction ($D = 0°$, $I = 23.2°$) in the study area. A minor decrease in the precision parameter after the tilt-correction indicated a negative fold test (McFadden, 1990) at the 95% confidence level. However, the classical fold test of McElhinny (1964) proved inconclusive at the 95% and 99% levels. The fold test of McFadden (1990) also proved inconclusive at the 99% level. Because the bedding attitudes at 11 of the sampling sites had very shallow dipping (1° to 14°), the applied fold tests cannot be considered sufficiently effect to infer a pre-folding origin for the high-temperature remanent component.

3.3.2. The NRM directions of the Type-B samples

In this category of samples, normal polarity directions were observed in 7 sites in the Sihanoukville area and 3 sites in the Koah Kong area. The samples in the remaining 7 sites in the Sihanoukville area and 1 site in the Koah Kong area revealed dual polarity directions (Fig. 3d). Reversed polarity directions were observed in 21 samples taken from 9 sites (Table 1). After inverting the reversed directions into normal, the site mean directions of the type-B samples revealed a wide range of precision parameters (between 5 and 475). The formation mean direction calculated on the basis of 19 site was $D_g = 5.2°$, $I_g = 18.5°$, $k_g = 116.1$ and $\alpha_{95} = 3.1°$ in geographic coordinates and $D_s = 4.3°$, $I_s = 14.7°$, $k_s = 44.7$ and $\alpha_{95} = 5.1°$ in stratigraphic coordinates (Fig. 7). The fold tests of McElhinny (1964) and McFadden (1990) proved negative at the 95% and 99% confidence levels, respectively. Because of its statistical significance, the formation mean direction in ge-

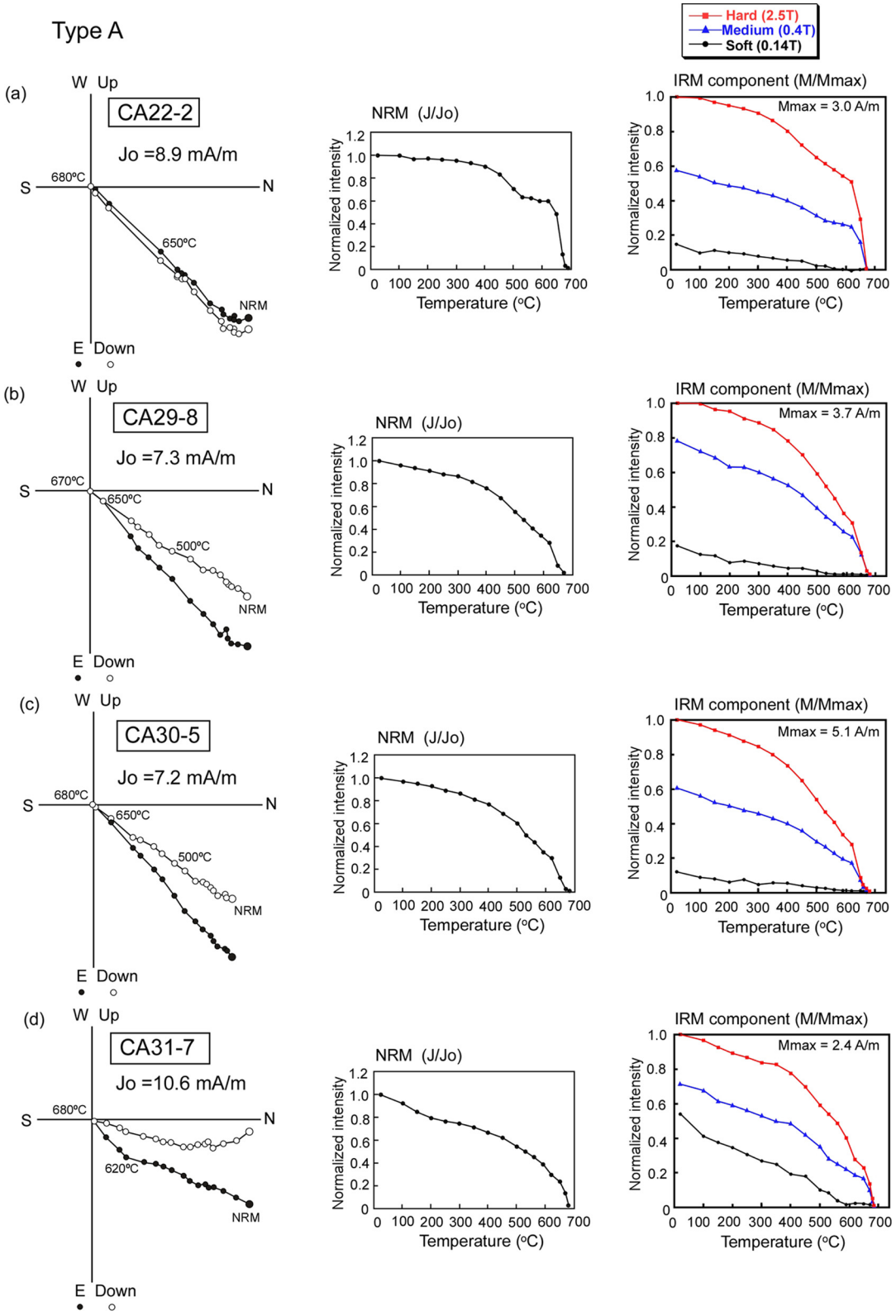


Fig. 4. Thermal demagnetization results of Type-A samples on Zijderveld diagrams together with the NRM intensity decay curves. The thermal demagnetization results of the composite IRMs imparted at DC fields of 2.5 T, 0.4 T and 0.14 T along *z*, *y* and *x* axes of the specimens, respectively are also shown on the adjacent plot.

ographic coordinates was recognized as the characteristic direction for the Type-B samples. This direction is almost parallel to the present IGRF and axial geocentric dipole in the study area, which indicates its post-folding secondary acquisition in the recent past.

4. Rock magnetism

A progressive acquisition of isothermal remanent magnetization (IRM) up to a maximum field of 2.5 T was performed on the representative samples from the Type-A, Type-B and Type-A+B cate-

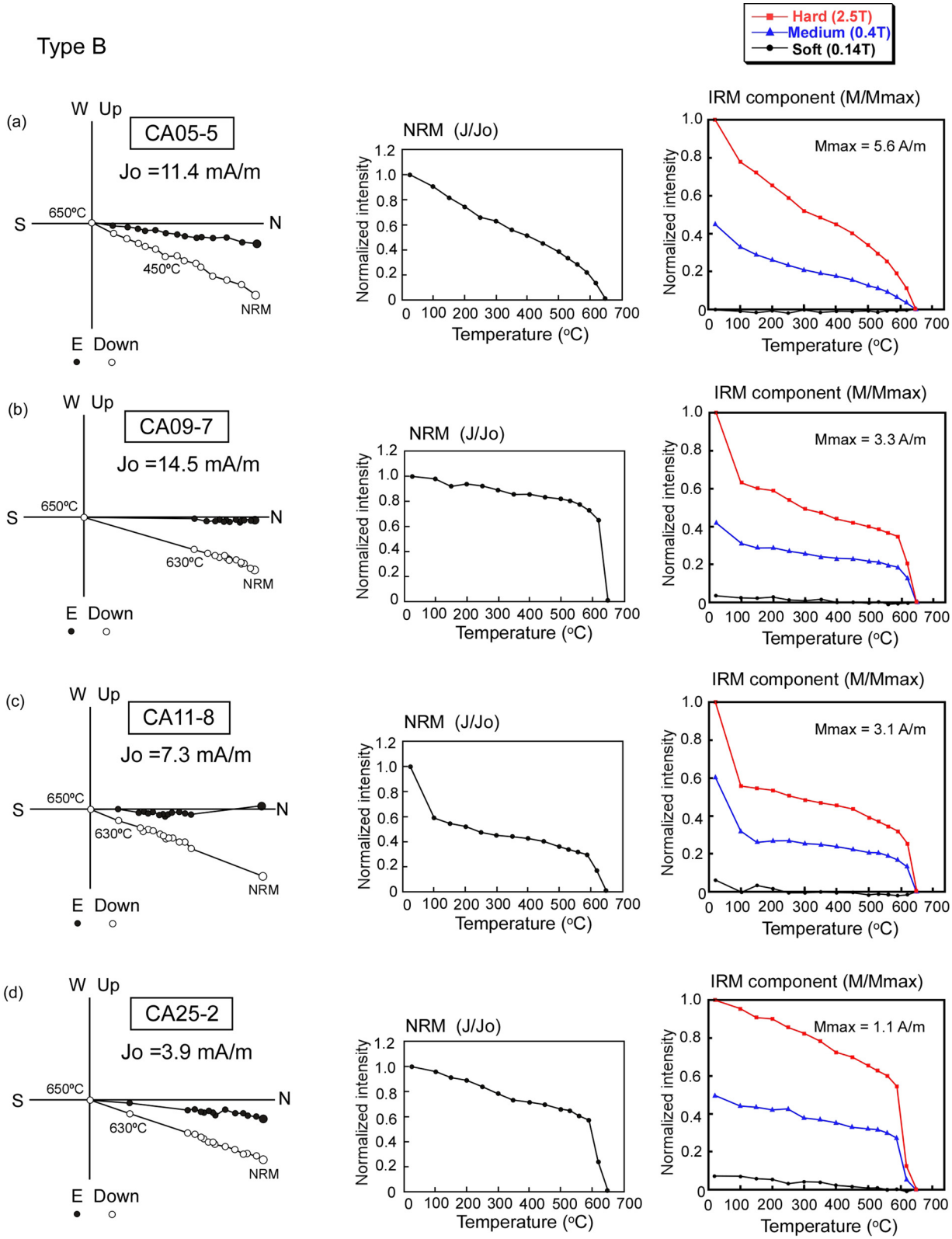


Fig. 5. Thermal demagnetization results of Type-B samples on Zijderveld diagrams together with the NRM intensity decay curves. The thermal demagnetization of the composite IRMs imparted at DC fields of 2.5 T, 0.4 T and 0.14 T along *z*, *y* and *x* axes of the specimens, respectively are also shown on the adjacent plot.

gories using a 2 G Enterprises 2G660 Pulse Magnetizer. The same specimens were then subjected to back-field IRM up to a maximum field of 1000 mT. A thermal demagnetization (ThD) of the composite IRMs (imparted at 2.5 T, 0.4 T and 0.14 T along *z*, *y* and *x* axes of the specimens, respectively) was conducted to identify the unblocking temperature spectra of the representative samples (Lowrie, 1990). The applied IRMs were measured using a Natsuhara Giken SMM-85 spinner magnetometer.

Although the IRM acquisition curves for the analyzed 30 samples indicated no saturation up to a maximum applied field of

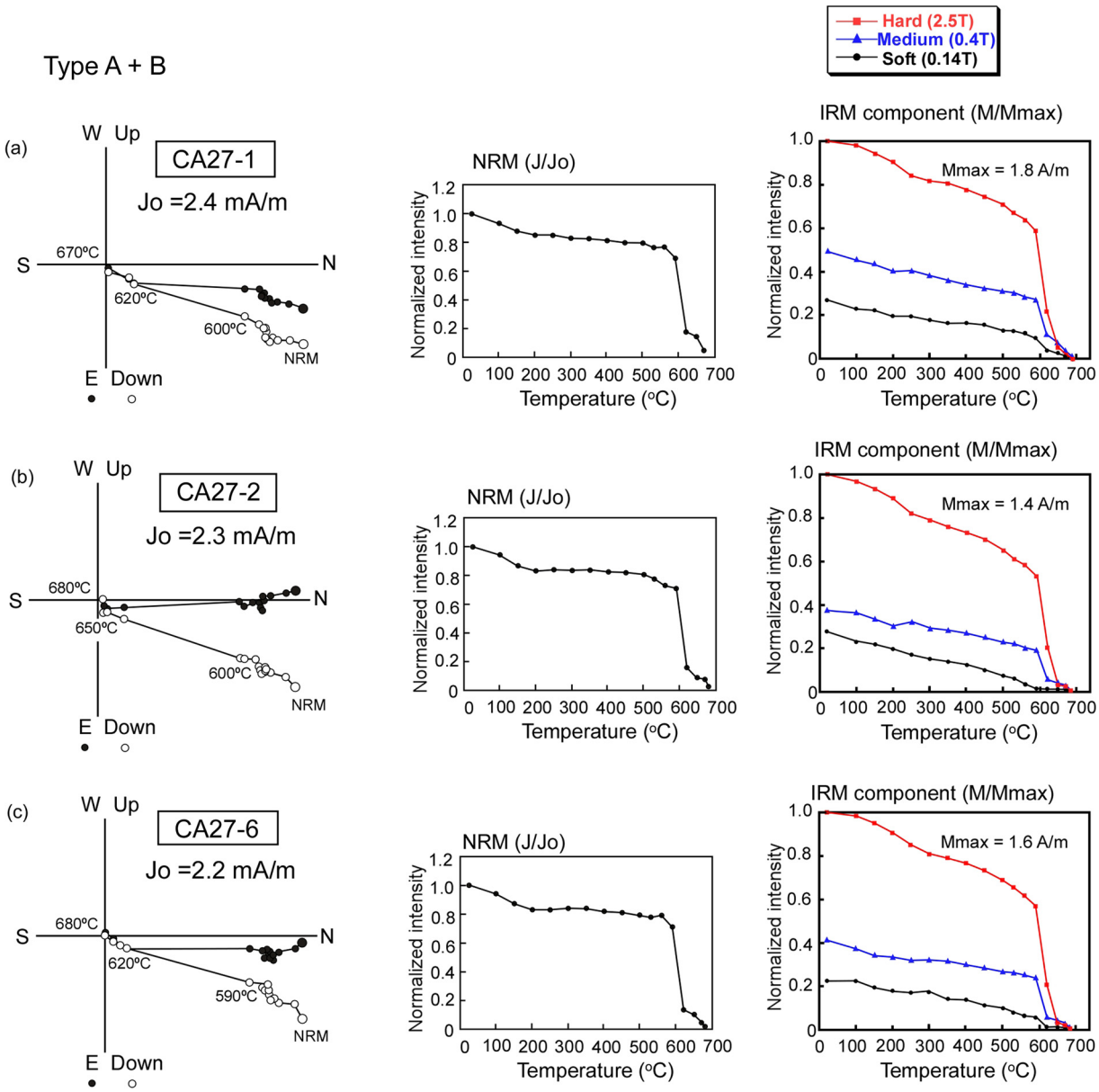


Fig. 6. Thermal demagnetization results of the samples from site CA27 (Type-A+B) on Zijderveld diagrams together with the NRM intensity decay curves. The thermal demagnetization of the composite IRMs imparted at DC fields of 2.5 T, 0.4 T and 0.14 T along *z*, *y* and *x* axes of the specimens, respectively axes are also shown. Two unblocking temperature zones, 620–650 °C and 680–690 °C, were observed in the hard and medium IRM components.

2.5 T, their representative back-field demagnetization curves show remanence coercivities between 400 and 1000 mT, which reveals the presence of hematite or goethite as high-coercivity magnetic minerals. During the ThD procedure an abrupt drop in the intensity of the hard IRM component at approximately 100 °C confirmed the presence of goethite (Figs. 5b, c). Because of the very strong intensity, the composites IRMs acquired in several of the samples could only be measured by the spinner magnetometer after the 100 °C thermal demagnetization.

Nearly similar unblocking temperature spectra were observed during the thermal treatments of the NRMs and IRMs. In case of the IRM, an unblocking temperature range of 670 to 680 °C was observed in the hard and medium components of the Type-A samples (Fig. 4), whereas an unblocking temperature of 650 °C was observed in the Type-B samples (Fig. 5). Samples in the Type-A+B category (site CA27) revealed two unblocking temperature ranges of 620–650 °C and 680–690 °C in the hard and medium IRM components. Almost similar unblocking temperatures were observed in the NRM curves for the same samples during their thermal demagnetization (Figs. 4, 5).

The presence of magnetite is suspected in several of the samples from sites CA12 and CA26, where an unblocking temperature of 580 °C appeared in the soft coercivity fraction (0.14 T) of the IRM demagnetization curves. The low-temperature NRM component unblocked at 560–590 °C during the ThD procedure also confirmed the presence of magnetite.

5. Microscopic observation

To study the occurrence of iron oxides in the sampled rocks and their sedimentary textures, petrographic observations were performed on 6 representative samples from all three types (Type-A; CA22-2 and CA30-8, Type-A+B; CA27-6, Type-B; CA8-3, CA9-1, and CA26-4), using a Nikon optical microscope (Fig. 8). The observations of the polished thin sections under reflected light showed sporadically distributed hematite grains up to tens of micrometers

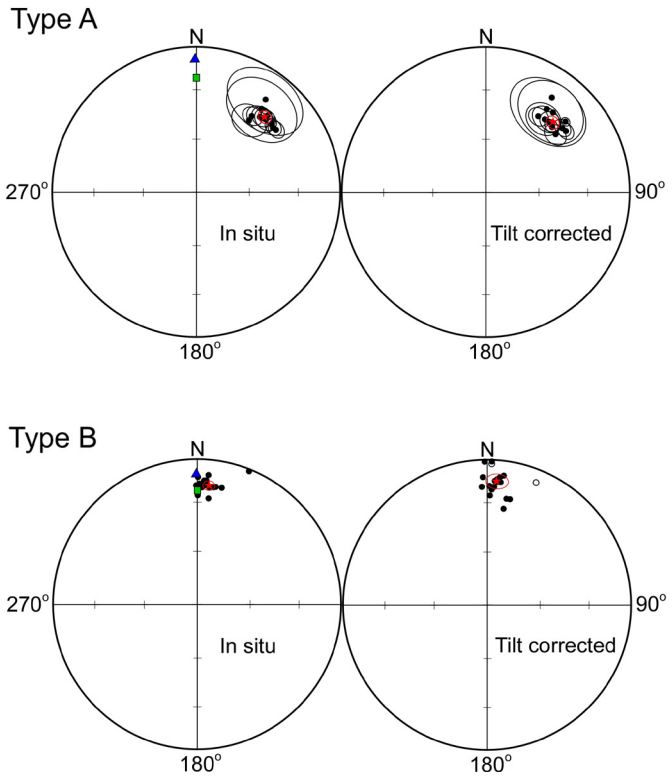


Fig. 7. Paleomagnetic directions of the samples from the Sihanoukville and Koah Kong areas. (A) Equal-area projections of the site mean directions (before and after tilt correction) for the Type-A sites with a 95% confidence limit. The solid (open) symbols indicate the projections onto the lower (upper) hemisphere. (B) Equal-area projections of the site mean directions (before and after tilt correction) for the Type-B sites. Red stars with an oval of the 95% confidence limit show the mean directions in both these figures. The green square and blue triangle indicate the present day geomagnetic field direction (IGRF) and geocentric dipole direction in the study area, respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

in all of the samples. The hematite grains under polarized light showed a distinctive reflection pleochroism, which revealed the presence of an ilmenite–hematite solid solution series (i.e., specular hematite). Those grains along with quartz, feldspar and lithic fragments constitute detrital ingredients of the studied red beds.

Based on the petrographic observations of the representative samples, two categories of sedimentary texture on the basis of grain-to-matrix ratio were recognized. On the basis of the defined unblocking temperature spectrum of the remanent magnetization, each of the two categories could be compared to either the Type-A and Type-A+B samples or to the Type-B samples. The quartz grains in the Type-A (CA22-2 and CA30-8) and Type-A+B (CA27-6) samples were up to ∼50 μm in size and had an angular to sub-angular morphology (Fig. 8I, II), which revealed a lower grain-to-matrix ratio. The matrix here was connected by proper networking, and the sections under polarized reflected light showed distribution of hematite pigment throughout. On the other hand, quartz grains in the Type-B samples (CA8-3, CA9-1, and CA26-4) were up to ∼200 μm in size, and had low sphericity and angular to sub-angular morphologies. The concave-convex shaped features and sutured contacts of these grains caused very high grain-to-matrix ratios in the Type-B samples. Under the polarized reflected light, thin coatings of hematite pigment were observed around grains (pore-filling hematite pigment).

The occurrence of magnetite in the studied samples was identified by rock magnetic investigation. Bearing in mind those results, we observed the representative samples under polarized light to identify the presence of magnetite particles, which could be distinguished from the hematite grains by the absence of reflection pleochroism. However, identification of the magnetite particles under optical microscope proved difficult because of their small grain sizes.

6. Discussion

6.1. Primary and secondary magnetizations in the Middle Jurassic–Early Cretaceous red beds

The NRMs in the Type-B samples were secondary in origin because the fold tests were negative, and the corresponding in-situ formation mean direction was sub-parallel to the axial geocentric dipole direction in the study area ($D = 0°$, $I = 23.2°$). Twenty one of 135 samples preserved a record of the reversed polarity directions, which suggests an acquisition of the secondary magnetization during the Matuyama and Brunhes chrons. As evident from the thermal demagnetization of the NRMs and composite IRMs, the remanence in the samples was carried by magnetic minerals with unblocking temperatures ranging from 650 to 670 °C.

The NRMs in the Type-A samples were probably primary in origin because the unblocking temperatures of those samples varied between 670 and 690 °C, which were clearly higher than those of the Type-B samples. The unblocking temperature is additional diagnostic to determine the primary or secondary origin of NRMs in red beds. Some researchers (Lee et al., 1996; Cox et al., 2005; Tan et al., 2007) have reported two types of NRM components in the red sandstones from the Korean Peninsula and North America, i.e, a low-temperature component (LTC) with an unblocking temperature of 610–660 °C and a high-temperature component (HTC) with unblocking temperature of 680–690 °C. Secondary and primary origins were been assigned to the LTC and HTC, respectively. They ascribed the LTC to secondarily acquired chemical remanent magnetization (CRM) by authigenic pigmentary hematite and/or a newly formed maghemite. On the other hand, specular hematite, which carries primary remanent magnetization (determined as HTC), was linked to early diagenetic post-depositional or detrital remanent magnetization. Liu et al. (2011) and Kodama (2012) suggested a close link between the unblocking temperature of the remanence components and the origin of the magnetic minerals. This close link has been experimentally confirmed (Jiang et al., 2015). Therefore, the NRMs with unblocking temperatures in the range 680–690 °C in the Type-A samples were ascribable to primary magnetization carried by specular hematite.

Our microscopic observation showed that specular hematite was present in almost all of the samples rather than martite as a constituent magnetic mineral. Although no perceptible difference was observed in the magnetic minerals between the Type-A and Type-B samples, hematite pigments were distributed throughout the matrix in the Type-A samples, whereas pore-filling hematite pigments were observed in the Type-B samples as a cementing material. The presence of hematite in the interstices of quartz grains have also been observed in the remagnetized Cambrian and Triassic rocks of the Okcho'n zone on the Korean Peninsula (Otofuji et al., 1989). Pore-filling hematite pigments thus played an important role in the acquisition of the secondary NRM. Considering the observations of Liu et al. (2011) and Kodama (2012), the NRMs of the Type-A and Type-B samples in this study were probably carried by specular and pigmentary hematite, respectively.

Although the origin of the magnetic minerals in the Type-B samples (that carried an unblocking temperature of 650 °C) remains unclear, the presence of goethite provides a clue regarding its secondary nature. Although Fan et al. (2006) observed a thermal transformation of goethite to hematite during laboratory experiments, the natural dehydration of goethite at ambient temperatures over thousands of years has also been reported

Table 1
Characteristic remanent magnetization direction.

Site	Location		Strike (°)	Dip (°)	NRM intensity	100 °C ThD	Goethite	n/N	Type	Porality (N/R)	In-situ		Tilt-corrected		k	α_{95} (°)
	Lat. (°N)	Lon. (°E)			mA/m	mA/m					D_g (°)	I_g (°)	D_s (°)	I_s (°)		
Sihanoukville																
CA05	10°42′53.7″	103°47′53.4″	228.4	10	8.3 ± 2.7	7.4 ± 2.8		11/12	B	11/0	0.1	23.3	357.8	15.7	7.3	18.2
CA06	10°42′55.5″	103°47′52.7″	62.4	4	17.8 ± 14.5	17.1 ± 14.5		9/10	B	9/0	5.3	11.9	6	15	6.6	21.7
CA07	10°43′32.9″	103°47′50.7″	110.4	14				0/0								
CA08	10°43′33.4″	103°47′49.6″	170.4	11	16.7 ± 10.8	14.5 ± 8.7	G	9/9	B	9/0	3.4	15.6	0.1	17.8	61.3	6.6
CA09	10°43′33.8″	103°47′49.8″	316.4	5	4.8 ± 4.2	4.3 ± 4.2	G	7/8	B	7/0	1.7	18.2	2.7	14.6	475.1	2.8
CA10	10°34′36.2″	103°38′29.4″	358.4	11	48.6 ± 49.4	11.2 ± 9.3	G	9/11	B	9/0	6.7	27.7	11.9	25.6	17.6	12.6
CA11	10°34′35.5″	103°38′29.3″	358.4	11	20.9 ± 34.6	8.0 ± 6.4	G	10/10	B	10/0	359.6	19.4	3	18.8	161	3.8
CA12	10°36′10.1″	103°32′52.4″	198.4	10	1.4 ± 1.1	1.0 ± 0.81	G	5/6	B	4/1	6	27.2	1.4	24.7	10.2	25.2
CA13	10°36′10.1″	103°32′52.4″	70.4	8	3.8 ± 2.8	3.0 ± 2.1	G	5/5	B	4/1	8.6	18.8	10.2	25.7	85.2	8.3
CA14	10°36′10.1″	103°32′52.4″	238.4	18	35.9 ± 65.0	1.6 ± 0.93	G	7/8	B	6/1	21.1	1.7	22.1	−9.2	5	29.9
CA15	10°36′10.8″	103°32′52.5″	234.4	10	62.7 ± 57.1	13.0 ± 12.6	G	8/8	B	7/1	7.2	19.7	5.2	12.3	79.3	6.3
CA16	10°36′10.8″	103°32′52.5″	231.4	29	6.4 ± 8.3	3.3 ± 4.0	G	5/7	B	3/2	4.8	19.7	1.7	−2.1	22.5	16.5
CA17	10°36′10.8″	103°32′52.5″	205.4	16	42.9 ± 71.4	5.9 ± 3.7	G	8/9	B	3/5	182.4	−19.2	178.2	−12.4	80.6	6.2
CA18	10°36′05.9″	103°32′50.0″	200.4	27				0/0								
CA19	10°36′05.9″	103°32′50.0″	230.4	21	16.9 ± 11.1	12.3 ± 12.7	G	8/8	B	8/0	4.1	16.1	2.1	0.6	16.7	14
CA20	10°36′05.9″	103°32′50.0″	213.4	21	3.4 ± 2.2	1.8 ± 0.78	G	6/10	B	1/5	192.4	−19.1	187.2	−10.6	97.6	6.8
CA21	10°36′05.9″	103°32′50.0″	221.4	19	78.8 ± 128	5.1 ± 2.6	G	8/9	B	5/3	0.4	13	358.8	0.3	40.8	8.8
Koah Kong																
CA22	11°34′43.0″	103°08′53.0″	111.5	2	6.3 ± 2.8	6.3 ± 2.7		8/8	A	8/0	41.1	31.8	41.6	33.7	115	5.2
CA23	11°34′42.8″	103°08′54.0″	97.5	7	3.4 ± 1.1	3.3 ± 1.1		5/6	A	5/0	42.7	29.1	45.3	34.7	346.4	4.1
CA24	11°34′14.6″	103°08′00.2″	70.5	14	52.6 ± 42.1	0.92 ± 0.33	G	8/9	B	6/2	6.6	20.2	9.9	32.6	86.5	6
CA25	11°34′25.3″	103°05′27.0″	175.5	7	4.4 ± 3.4	4.3 ± 3.4		7/7	B	7/0	4.8	19.2	2.3	20.2	83.1	6.7
CA26	11°34′25.3″	103°05′27.0″	175.5	7	5.8 ± 12.7	5.4 ± 1.2		5/8	B	5/0	3.8	18	1.6	18.9	190.6	5.6
CA27	11°35′12.7″	103°05′11.2″	41.5	3	2.9 ± 0.93	2.8 ± 0.93		10/10	A+B, B							
								4/10	B	4/0	359.8	19.4	0.6	21.4	831.4	3.2
								6/10	A	6/0	38.5	27.1	40	39.5	13.6	18.8
CA28	11°36′14.9″	103°03′36.5″	8.5	1	8.8 ± 2.0	8.5 ± 1.9		9/9	A	9/0	49.8	30.9	50.3	30.2	119.4	4.7
CA29	11°36′14.9″	103°03′36.5″	8.5	1	6.7 ± 0.91	6.6 ± 0.84		9/9	A	9/0	52.1	30.3	52.5	29.6	123.5	4.7
CA30	11°36′14.9″	103°03′36.5″	322.5	4	7.2 ± 1.1	7.0 ± 1.1		9/9	A	9/0	47.8	29.7	48	25.7	447.7	2.5
CA31	11°36′16.8″	103°03′22.2″	136.5	7	9.3 ± 2.4	8.5 ± 1.7		4/4	A	4/0	37.1	21.2	36.6	28.1	22.7	19.7
CA32	11°36′21.9″	103°03′15.7″	348.5	4	2.7 ± 0.78	2.5 ± 0.75		8/8	A	8/0	36.5	37.9	38.4	34.9	34.6	9.6
CA33	11°36′22.7″	103°02′31.0″	210.5	3	6.2 ± 1.1	6.0 ± 0.98		9/9	A	9/0	36.6	35	34.5	35.3	97.6	5.2
CA34	11°36′22.7″	103°02′31.0″	210.5	3	10.0 ± 2.1	9.8 ± 1.9		7/7	A	7/0	41.4	33.2	39.4	33.7	68.1	7.4
CA35	11°36′21.3″	103°02′28.9″	242.5	2				0/0								
CA36	11°36′23.7″	103°02′18.6″	253.5	11				0/0								
CA37	11°36′24.6″	103°02′21.8″	74.5	11	10.8 ± 3.3	10.7 ± 3.3		6/6	A	6/0	43.8	31.1	50.3	36.2	120.3	6.1
B Type mean direction								19/29			5.2	18.5	–	–	116.1	3.1
											–	–	4.3	14.7	44.7	5.1
A Type mean direction								11/29			42.6	30.8	–	–	162	3.6
											–	–	43.4	31.9	158.7	3.6
								N			Lat. (°N)		Lon. (°E)			A_{95} (°)
VGP (HTC)	11.6°	103.1°						11			47.7		178.9			3.6

The paleomagnetic directions of the high-temperature components are listed in this table. 100 °C ThD: NRM intensity after a 100 °C thermal demagnetization. G: the presence of goethite is indicated by the abrupt decrease in the intensities approximately 100 °C. N and n: the number of samples measured and used for the calculations, respectively. As explained in the text, the unblocking temperatures were categorized into three types, i.e., Type-A (670–690 °C), Type-B (620–670 °C) and Type-A+B (620–650 °C and 670–690 °C). Because site CA27 contained both types of samples (A+B and B), two site mean directions were calculated. D (D_s) and I_g (I_s) are the declination and inclination, respectively. k is the Fisherian precision parameter (Fisher, 1953), while α_{95} is the radius of a cone at the 95% confidence level about the mean direction. CA07, CA18, CA35 and CA36: samples from these sites were discarded because of their fragile nature.

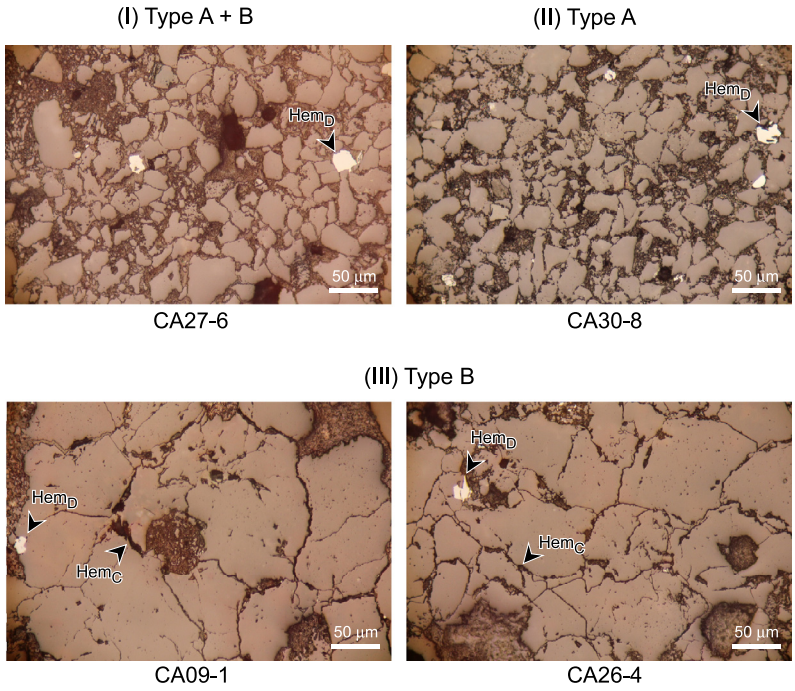


Fig. 8. Reflected-light photomicrographs of the representative samples. (I) Type-A+B (CA27-6), (II) Type-A (CA30-8) and (III) Type-B (CA09-1 and CA26-4). Specular hematite was identified from all of the types. Hematite pigments are distributed throughout the network of the matrix in the Type-A sample, whereas thin coatings of hematite pigment around grains and within pores were observed in the sandstones cement in the Type-B samples. Hem_D and Hem_C denote detrital hematite and hematite coating (pigment), respectively.

(Ozden and Dunlop, 2000). In addition, Gualtieri and Venturelli (1999) reported the formation of an intermediate phase hematite with nonstoichiometric composition ("*protohematite*") as a result of goethite decomposition. The unblocking temperature range of "*protohematite*" may be lower than that of the specular hematite. A similar trend of a low unblocking temperature (550 °C) has been observed in magnetite, based on thermomagnetic analyses of the goethite rich Franconian limestone.

Bearing in mind the rock- and paleomagnetic information discussed above, we conclude that the tilt corrected mean direction from 11 of the Type-A sites ($D_s = 43.4°$, $I_s = 31.9°$, $k_s = 158.7$ and $\alpha_{95} = 3.6°$) is the Late Jurassic to Early Cretaceous characteristic direction for the Phuquoc Formation in the Koah Kong area (11.6°N, 103.1°E). The corresponding paleomangetic pole for this direction is located at 47.7°N, 178.9°E ($A_{95} = 3.6°$).

6.2. Uniform southward displacement of the Indochina Peninsula

The Late Jurassic–Early Cretaceous characteristic direction ($D_s = 43.4°$ and $I_s = 31.9°$) obtained from the 11 Type-A sites indicates an easterly deflected declination and a fairly steep inclination. The displacement in the study area was estimated by comparing the observed paleolatitude with that expected from the 130 Ma East Asian APWP (Cogné et al., 2013). For this purpose, paleolatitudes of 17.3°N and 11.3°N were calculated from the observed and expected inclinations ($I_{ob} = 31.9°$ and $I_{exp} = 21.7°$), which indicate a 6.0° ± 3.5° southward displacement of the Koah Kong area. The projection in Fig. 9 shows the calculated displacement between the Koah Kong pole (47.7°N, 178.9°E and $A_{95} = 3.6°$) and the 130 Ma East Asian pole (79.9°N, 193.5°E and $A_{95} = 2.6°$). This amount is concordant with the post-Cretaceous displacement of 7.1° ± 2.2° estimated for the composite Shan-Thai–Indochina Block (Takemoto et al., 2009). We thus predict a southward displacement of a composite block that included the Koah Kong and consisted of both Shan-Thai and Indochina blocks.

The Khorat Basin (16.7°N, 103.9°E) is located in the central part of Indochina Block and is believed to be connected to the

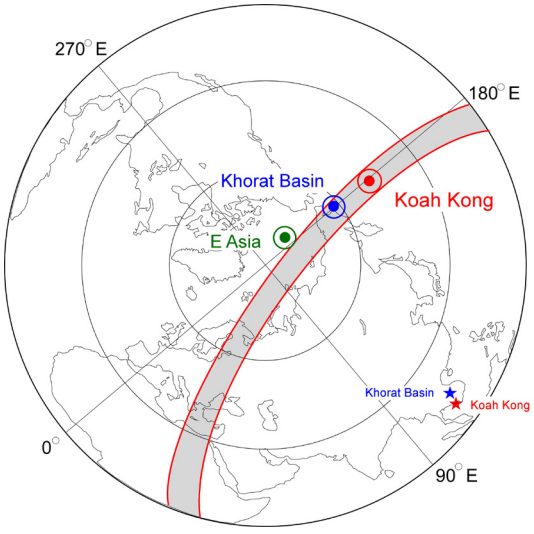


Fig. 9. Paleomagnetic pole positions for the Koah Kong area (red), Khorat Basin (blue) and East Asia (green) with 95% confidence limits. The grey stripe with red borders indicates the 95% confidence limit for the Koah Kong area pole (this study) centered on its current sampling location. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Phuquoc–Kampot Som Basin (Morley, 2013). We calculated a Late Jurassic to Cretaceous paleomagnetic pole (61.8°N, 180.4°E with $A_{95} = 3.5°$) for the Khorat Basin using 11 data sets (Table 2) and compared it with the herein presented Late Jurassic–Early Cretaceous paleomagnetic pole of the Koah Kong area. The paleomagnetic pole from the Khorat Basin falls on the 95% confidence small-circle strip that passes through the paleomagnetic pole of the Koah Kong area (with A_{95} centered on the Koah Kong area) (Fig. 9), which indicates that no relative latitudinal displacement has occurred between two areas. This result is quantitatively confirmed by the differences in the paleolatitudes and present day

Table 2
Jurassic to Cretaceous paleomagnetic results from the Indochina Peninsula (the Shan-Thai and Indochina blocks).

Locality			Age	*N*	Rock type	Fold test	Observed direction			Reference
Name	Lat. (°N)	Lon. (°E)					Dec. (°)	Inc. (°)	a_{95} (°)	
Shan-Thai Block										
Yunlong	25.8	99.4	K2	20	Redbeds	Positive	40.2	49.9	3.9	Sato et al. (1999)
Yunlong	25.8	99.4	K2	29	Redbeds	Positive	38.3	50.7	3.4	Yang et al. (2001)
Xiaguan	25.6	100.2	K2	9	Redbeds	Positive	6.9	47.7	8.6	Huang and Opdyke (1993)
Weishan	25.4	100.2	J3	5	Redbeds	Inconclusive	7.3	25.3	10.4	Huang and Opdyke (1993)
Yongping	25.5	99.5	K1	12	Redbeds	Positive	42.0	51.1	15.7	Funahara et al. (1993)
Jingdong	24.5	100.8	K1-2	13	Redbeds	Positive	8.3	48.8	7.7	Tanaka et al. (2008)
Luxi	24.3	98.4	J2	6	Redbeds	Inconclusive	99.7	35.2	11.3	Huang and Opdyke (1993)
Zhengyuan	24.1	101.1	K1-2	7	Redbeds	Positive	61.8	46.1	8.1	Tanaka et al. (2008)
Zhengyuan	24.1	101.1	K1-2	4	Redbeds	Positive	144.2	49.4	6.4	Tanaka et al. (2008)
Jinggu	23.6	100.5	J2	10	Redbeds	Inconclusive	83.3	36.8	5.4	Huang and Opdyke (1993)
Jinggu	23.4	100.4	K1	–	Redbeds	Inconclusive	84.4	39.6	17.8	Chen et al. (1995)
Jinggu	23.4	100.5	K2	7	Redbeds	Positive	115.8	36.0	6.3	Chen et al. (1995)
Jinggu	23.4	100.9	K2	8	Redbeds	Inconclusive	79.4	43.3	9.1	Huang and Opdyke (1993)
Pu'er	23.0	101.0	K1-2	25	Redbeds	Positive	59.9	45.2	5.1	Sato et al. (2007)
Zhengwan	22.8	100.9	K1	11	Redbeds	Positive	51.8	47.9	6.9	Kondo et al. (2012)
Dadugang	22.4	101.0	K1	12	Redbeds	Positive	64.1	48.1	7.3	Kondo et al. (2012)
Menlun	21.9	101.2	K2	6	Redbeds	Positive	33.2	30.9	8.2	Tong et al. (2013)
Menlun	21.9	101.2	K1	19	Redbeds	Positive	45.3	45.9	11.0	Tong et al. (2013)
Mengban	21.8	101.6	K1-2	4	Redbeds	Positive	50.5	31.0	6.4	Tong et al. (2013)
Phong Saly	21.6	101.9	J3-K1	19	Redbeds	Positive	29.8	32.7	9.1	Takemoto et al. (2009)
South Mengla	21.4	101.6	K1-2	14	Redbeds	Positive	51.2	46.4	5.6	Tanaka et al. (2008)
Mengla	21.4	101.6	K1	14	Redbeds	Positive	46.9	42.2	7.7	Tong et al. (2013)
Kalaw	20.7	96.5	J-K	13	Redbeds	Positive	44.7	23.4	6.1	Richter and Fuller (1996)
Nan	19.2	101.0	J1-3	11	Redbeds	Positive	32.2	33.3	12.2	Aihara et al. (2007)
Indochina Block										
Lai Chau	22.3	103.4	K	5	Redbeds	Positive	12.2	40.1	4.7	Takemoto et al. (2005)
Yen Chau	21.0	104.4	K	8	Redbeds	Positive	3.2	26.7	12.9	Takemoto et al. (2005)
Khorato Basin										
Borikhanxay	18.5	103.8	J3-K1	18	Redbeds	Positive	41.8	46.6	8.0	Takemoto et al. (2009)
Borikhanxay	18.3	103.4	K2	16	Redbeds	Positive	21.2	37.1	7.7	Singsoupho et al. (2014)
Amphoe Bung Kuan	18.2	103.9	K2	14	Redbeds	Inconclusive	31.8	28.7	3.5	Charusiri et al. (2006)
Vientiane	18.1	102.4	k1	10	Redbeds	Positive	25.9	37.4	10.2	Singsoupho et al. (2014)
Khammounane+Savannaket	16.7	105.4	K1	8	Redbeds	Positive	28.8	45.7	5.0	Singsoupho et al. (2014)
Savannaket	16.5	105.5	K2	12	Redbeds	Positive	27.7	38.0	4.1	Singsoupho et al. (2014)
Nam Nao	16.5	103.0	K1	10	Redbeds	Inconclusive	28.1	40.5	2.4	Yang and Besse (1993)
Nam Nao	16.5	103.0	J3	10	Redbeds	Inconclusive	26.6	37.7	2.6	Yang and Besse (1993)
Muan Sakon Nakhon	16.5–17.2	102.5–104.1	K2	8	Redbeds	Inconclusive	31.4	27.1	9.4	Charusiri et al. (2006)
Muan Sakon Nakhon	16.5–17.2	102.5–104.1	K1	4	Redbeds	Positive	31.8	38.3	5.7	Charusiri et al. (2006)
Muan Phin	16.5	106.0	J (K1)	17	Redbeds	Syn-folding at K1	30.8	39.9	3.0	Takemoto et al. (2009)
Southern tip										
Da Lat	10.4–12.9	105.0–109.4	K	21	Ryolite, Ryolite tuff, granite, silt	None	14.5	33.3	6.3	Chi and Dorobek (2004)
Da Lat	11.7	108.3	K	21	Redbeds	Positive	11.4	35.4	1.7	Otofuji et al. (2012)
Koah Kong	11.6	103.1	J3-K1	11	Redbeds	Inconclusive	43.4	31.9	3.6	This study
East Asia (Cogné et al., 2013)										
130 Ma (pole position)	79.9°N	193.5°E	$\alpha_{95} = 2.6°$							

Lat. and Long. are latitude and longitude, respectively. Ages are: J for Jurassic, J1 for Lower Jurassic, J2 for Middle Jurassic, J3 for Upper Jurassic; K for Cretaceous, K1 for Lower Cretaceous; K2 for Upper Cretaceous, N is the number of sites used for the calculations. Dec. and Inc. are declination and inclination, respectively. α_{95} is the radius of a cone at the 95% confidence level about the mean direction.

latitudes between the Khorat Basin and Koah Kong, i.e., 5.1° ± 2.9° and 5.1°, respectively. In combination with the geological connection between the Khorat and the Phuquoc–Kampot Som basins, they formed a single sedimentary basin during the Mesozoic Era.

Although several geologic and tectonic maps of the Indochina Peninsula have been published (e.g., Fyhn et al., 2009; Moreley et al., 2013), a clear geological relationship between the Phuquoc–Kampot Som and Khorat basins has remained unresolved. In that regard, the most widely cited geological relationship is the reported extent of the Wang Chao Fault (Mae Ping shear zone), which runs from the Sagain Fault Zone to the vicinity of the Mekong delta (Briais et al., 1993; Huchon et al., 1994; Rangin et al., 1995; Leloup et al., 1995; Laccasin et al., 1997; Morley, 2002; Gilly et al., 2003; Lepvrier et al., 2004; Hirsch et al., 2006; Fyhn et al., 2009; Sone et al., 2012). Some recent studies have suggested that the Wang Chao Fault (with splay geometry) passes between the Phuquoc–Kampot Som and Khorat basins but disappears before reaching the Mekong delta (Morley et al., 2007, 2013; Fyhn et al., 2010; Ridd and Morley, 2011; Palin et al., 2013). The consistency in the southward displacement between the Phuquoc–Kampot Som and Khorat basins implies that there has been no relative tectonic movement along the Wang Chao Fault. Therefore, we conclude that the Wang Chao Fault played no role in the southward displacement of the Shan-Thai and Indochina blocks within the framework of the Indochina Peninsula.

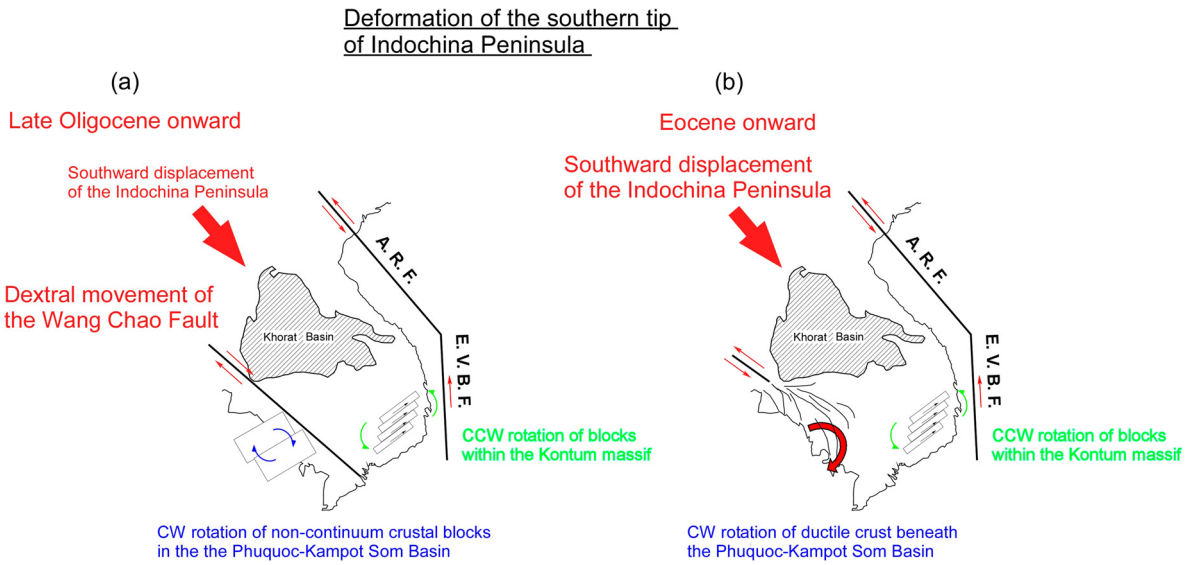


Fig. 10. A model explaining the deformation in the southern tip of the Indochina Block. (a) CW rotation of the non-continuum crust under the Koah Kong area. From the latest Oligocene onward, the Indochina and Shan-Thai blocks underwent a coherent southward displacement along the Red River Fault. Right-lateral movement along the Wang Chao Fault from the Late Oligocene onward is responsible for the CW rotation of the non-continuum crustal blocks that comprise the Phuquoc–Kampot Som Basin. As described by Otofuji et al. (2012), a CCW rotation in southeastern Vietnam occurred contemporarily as a result of a bookshelf type deformation in the non-continuum crust along the East Vietnam Boundary Fault (E.V.B.F). (b) CW rotation of the viscoplastic crust under the Koah Kong area. The crust beneath the Phuquoc–Kampot Som Basin behaved in viscoplastic manner during the southward displacement of the Indochina and Shan-Thai blocks since Eocene. A splaying geometry with a concave northeast attitude in the Mae Ping Fault Zone (a part of the Wang Chao Fault) is one of the visible traces of the CW rotation of the viscoplastic crust. (a, b) When the Indochina and Shan-Thai blocks moved southward after the collision of India with Asia in the Eocene, the lithosphere beneath the southern tip of the Indochina Block behaved in non-rigid manner and moved heterogeneously. In contrast, the rigid lithosphere beneath the Khorat Basin rotated homogeneously.

6.3. Tectonic rotation model for the Phuquoc–Kampot Som Basin

An easterly deflected declination ($D_s = 43.4°$) is another characteristic of the Late Jurassic to Early Cretaceous paleomagnetic direction in the study area (Koah Kong). When compared with the expected declination from the 130 Ma East Asian pole (Cogné et al., 2013), a CW rotation of 33.1° ± 4.0° about a vertical axis is calculated for the Koah Kong area. The predominant easterly deflected declinations in the Jurassic–Cretaceous rocks of the Shan-Thai and Indochina blocks (Fig. 1) suggest an occurrence of a CW rotation in those blocks. The tectonic rotation map of Takemoto et al. (2009) also showed a CW rotational motion of more than 10° for the Indochina and Shan-Thai blocks as a composite tectonic unit after the collision of India with Asia. We therefore predict a CW tectonic rotation and southward displacement of the Koah Kong area as a part of the Shan-Thai–Indochina composite block during the extrusion of SE Asia after the gigantic India–Asia collision.

The most striking observation by this study from the Koah Kong area is a large CW rotation (33.1° ± 4.0°), which is 14.7° ± 5.2° larger than that reported from the Khorat Basin. This indicates that relative rotation occurred between the two basins. The rotation of a block about its vertical axis is generally explained by the non-continuum nature of the crust, assuming that the crustal block is undeformed and that the strain is accommodated by slip along faults (Fig. 10a). As described by MacDonald (1980), lateral motion along a boundary fault is one mechanism to rotate blocks. A strike-slip motion along the Wang Chao Fault probably played an important role in the large CW rotation of the Koah Kong area. Bearing in mind those explanations, we believe that the Wang Chao Fault is a boundary fault between the Phuquoc–Kampot Som and Khorat basins (Fig. 10a). The Wang Chao Fault remained under the influence of a left-lateral movement in the mid-Oligocene but the sense of motion changed to right-lateral in the latest Oligocene and continued onward (Lacassin et al., 1997; Morley, 2002, 2007; Fyhn et al., 2009). The right-lateral movement along the Wang Chao Fault might have been responsible for the CW rotation of the adjacent blocks that in turn formed the Phuquoc–Kampot Som Basin (MacDonald, 1980). On a regional scale, the southward displacement of the Indochina Peninsula facilitated this rotation under the regional compressive regimes. When cracks appeared in the displacing peninsula as a result of activation of the Wang Chao Fault, the regional compression became concentrated along it. Those compressional regimes together with the right-lateral shearing along the Wang Chao Fault caused the tectonic blocks in the south to rotate CW about vertical axis.

An alternative model for the CW rotation in the study area (Fig. 10b) can be built on the basis of the continuum nature of the crust in the Phuquoc–Kampot Som Basin during deformation. A splaying geometry and concave northeast attitude of the Mae Ping Fault Zone (a part of Wang Chao Fault) around the Thai–Cambodia border are some of the visible traces of crustal deformation in the area. Herein, the regional compression is considered to be the most likely cause of the tectonic deformation and the associated CW rotation of the viscoplastic crust. The splay geometry along the Mae Ping Fault is a possible expression of cracks in the fragile uppermost part of the viscoplastic crust. The CW tectonic rotation of the viscoplastic crust played a major role in absorbing regional compressions in the Indochina Block during its southward displacement after the India–Asia collision.

6.4. Deformation of the lithosphere below the southern tip of the Indochina Block

The southern tip of the Indochina Block was subjected to variable block rotations that caused internal tectonic deformations in the area. As mentioned in the above sections, the Koah Kong area was subjected to a CW rotation of 33.1° ± 4.0°, but the Da Lat area (11.7°N, 108.3°E) in southern Vietnam experienced only a 0.9° ± 3.2° rotation (Otofuji et al., 2012). A non-rigid and heterogeneous movement of the lithosphere is one response to internal deformation (Fig. 10). On the basis of seismic tomography (Lebedev and Nolet, 2003; Huang and Xu, 2011; Shapiro et al., 2008; Schaeffer and Lebedev, 2013; Yang et al., 2015), a low seismic velocity zone was identified at depths from 70 km to 150 km under

the Koah Kong (the Phuquoc–Kampot Som Basin) and Da Lat areas. These low velocity zones at comparable depths imply a non-rigid and viscoplastic nature of the lithosphere beneath the southern tip of the Indochina Block.

A correlation between the internal deformation and low seismic velocity zone is also observed in the Shan-Thai Block. Variable magnetic declinations in the Cretaceous rocks indicate the formation of an oroclinal bending, as visible in structural features of the Chongshan–Lancang–Chiang Mai belt (Tanaka et al., 2008) and in the Simao Arc (Kondo et al., 2012). The Shan-Thai Block was subjected to internal deformation as well as a CW rotation of the entire block. Seismic tomographic studies have revealed a low seismic velocity zone deep in the lithosphere under the Shan-Thai Block between 50 km and 160 km (Lebedev and Nolet, 2003; Shapiro et al., 2008; Li et al., 2006; Schaeffer and Lebedev, 2013; Yang et al., 2015).

In contrast to the above-mentioned two areas, a strong and rigid lithosphere prevails under the Khorat Basin (Fig. 10). The mean paleomagnetic direction reported for this basin shows a CW rotation of 19.5° ± 3.3° with respect to East Asia. The low standard deviation (3.3°) here indicates a small amount of variation in the rotation within this basin, which suggests that the Khorat Basin was uniformly subjected to the CW rotation as a coherent lithologic block. The uniform rotation of the Khorat Basin during its southward displacement implies the presence of a strong lithosphere beneath it. The rigid nature of the lithosphere up to a depth of 200 km under the basin has been confirmed by Rayleigh wave velocity tomography (Yang et al., 2015). We therefore conclude that a rigid lithosphere in the central part of the Indochina Block is bounded by its non-rigid counterparts in the Shan-Thai Block to the north and the southern tip of the Indochina Block to the south.

7. Conclusion

To detect tectonic deformations in the southern tip of Indochina Peninsula, we conducted a new paleomagnetic study on Late Jurassic to Early Cretaceous red beds of the Phuquoc Formation in the Phuquoc–Kampot Som Basin, Cambodia:

1. A high-temperature component (unblocked at 670–690 °C) in the red beds of Koah Kong area is identified as primary remanent magnetization. The tilt-corrected mean direction ($D_s = 43.4°$, $I_s = 31.9°$, $k_s = 158.7$, $\alpha_{95} = 3.6°$ and $N = 11$) of this component is considered to be the Late Jurassic to Early Cretaceous characteristic direction for the study area.
2. The Koah Kong area was subjected to a southward displacement of 6.0° ± 3.5° and a CW rotation of 33.1° ± 4.0° with respect to East Asia. The area was carried southward as a part of the Indochina Block.
3. The approximately 15° larger CW rotation of the Koah Kong area relative to the Khorat Basin was probably caused by dextral motion along the Wan Chao Fault since the mid-Oligocene.
4. The comparison between the Phuquoc–Kampot Som and Da Lat basins also indicates that a differential tectonic rotation occurred in the area, which implies that lithosphere is not rigid beneath the southern tip of the Indochina Block.

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